

Notes on scanned documents

A scanner does not read a page. It photographs it. The result is a grid of gray dots, and a PDF made from a scan is a container for those dots. There is no text inside such a file, only a picture of text.

A text layer is the part of a PDF that holds real characters. When a document is exported from a word processor, every letter is stored as a letter, and a program can copy it, search it and count it. A scan has no such layer unless someone adds one afterwards.

Optical character recognition, OCR for short, is the step that adds it. An OCR engine looks at the picture, guesses which shapes are letters, and writes the guesses back as text. The guesses are good on clean print and poor on faded ink.

For retrieval this matters more than it seems. A retriever can only find text it was given. If the parse of a scanned page returns nothing, the page is invisible to every question, however relevant it is.

What to check first

Before tuning anything else, open the parsed output and look for the page. An empty page in the parse means the file had no text layer for that page. Turning OCR on is the only fix; a better chunker cannot cut what is not there.

OCR is slow. Reading text from an image takes seconds per page, where reading a text layer takes milliseconds. Leave OCR off for documents that already have text, and turn it on when the parse comes back empty.

OCR is also imperfect. The letter l and the digit 1 are often confused, and a word split by a fold in the paper can come back as two words. A question that quotes a passage word for word may still miss when the passage was read by OCR.

The habit to build is simple. Parse first, read the parse, and only then decide what the pipeline needs. A scanned page tells you within a second that it needs OCR, if you look.